Suppose we want to learn a binary function $f(X)$ that classifies whether a picture contains a cat ($f(X) = 1$) or not ($f(X) = 0$). Our input dataset is composed of 250 images, each represented as an array of 1024 pixels. Say we split the data into a training set of 150 images and a test set of 100 images. Rather than using a machine learning algorithm to choose our hypothesis $h(X)$ that approximates the true function $f(X)$, we just decide to randomly pick hypotheses until we find a good one.

We'll say two hypotheses $g(X)$ and $h(X)$ are *distinct* if they provide a different labeling of our specific dataset. So if we only had one image, there's only two distinct hypotheses: either you label the image as 1 or 0.

a.) How many distinct hypotheses are there for the entire dataset of 250 images?

2^250

b.) If you randomly choose one such hypothesis, what is the probability that its labeling perfectly matches the 150 training images?

1/(2^150)

c.) Suppose we randomly choose a hypothesis and find out it does in fact perfectly label the 150 training images. What is the probability that it also perfectly labels the 100 test images?

1/(2^100)